

README — vendor-corpora/

Low-Level Systems Architecture: Qualcomm QAIRT/HTP & Google Android NDK/OS

Motto: *Xçineribus, in-variis-nunquam-varius, Novi-Æxentis-Copiæ, Vincent*

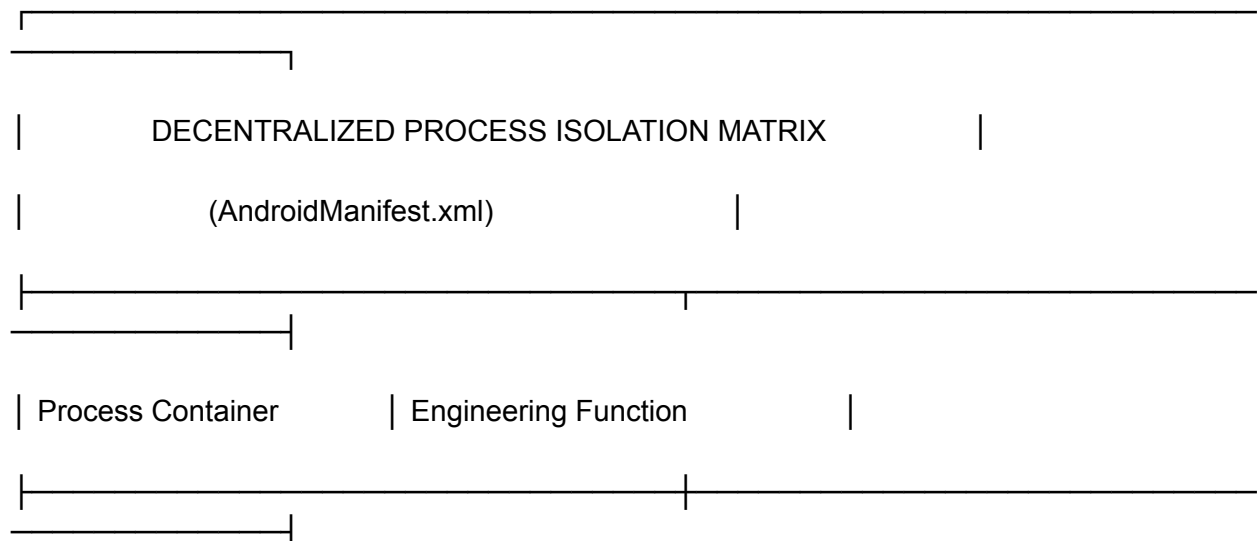
Scope: Definitive Technical Reference for Qualcomm Hexagon NPU Offloading and Google Android Bare-Metal OS IPC.

1. Architectural Mandate: Why This Library Is Decoupled

This corpus stores the immutable low-level engineering references governing how software interacts with mobile silicon. It is strictly partitioned into two specialized domains:

1. **Google Android Platform & NDK:** Process containerization, POSIX UNIX domain socket file descriptor passing (**SCM_RIGHTS**), zero-copy anonymous shared memory (**ASharedMemory**), and Low Memory Killer (LMK) evasion.
2. **Qualcomm QAIRT & Hexagon HTP SDK:** Hardware tensor pipelines, FastRPC kernel drivers, QNN execution providers, and GenieX on-device GGML Hexagon compilation-free inference.

2. Google Android Systems Architecture & NDK Subsystem



:orchestrator_daemon	Central coordination, WebSocket router	
:qairt_engine	Native Qualcomm HTP FastRPC driver	
:llamacpp_engine	GGML CPU/OpenCL fallback engine	

A. NDK Anonymous Shared Memory (ASharedMemory)

Standard Android Binder IPC is capped at **1MB total buffer size** across all transactions (`TransactionTooLargeException`). Passing multi-megabyte tensor weights, KV caches, or raw frame arrays over Binders causes instant runtime crashes.

- **The NDK Fix:** Memory pools are allocated natively via `ASharedMemory_create(name, size)`.
- **Zero-Copy Mapping:** The allocating daemon maps the region via `mmap( PROT_READ | PROT_WRITE, MAP_SHARED )`.
- **Kernel Protection Enforcement:** Before sharing, the daemon invokes `ASharedMemory_setProt(fd, PROT_READ)` to strip write permissions from child processes, guaranteeing that worker nodes cannot corrupt active model weights.

B. Linux Abstract Namespace UNIX Domain Sockets & SCM_RIGHTS

To bridge the shared memory file descriptor (`fd`) across isolated Linux processes without writing temporary files to physical flash storage (which degrades UFS 4.0 storage and spikes I/O wait times):

- **Abstract Namespace Binding:** Sockets are bound with `sun_path[0] = '\0'`, routing traffic entirely within the Linux kernel network namespace.
- **Ancillary Message Passing:** File descriptors are transmitted using `sendmsg()` and `recvmsg()` with POSIX control headers:

```
struct cmsghdr *cmsg = CMSG_FIRSTHDR(&msg);

cmsg->cmsg_level = SOL_SOCKET;

cmsg->cmsg_type = SCM_RIGHTS; // Direct kernel routing of native file descriptors

cmsg->cmsg_len = CMSG_LEN(sizeof(int));

*((int*)CMSG_DATA(cmsg)) = shared_mem_fd;
```

- **Zero Serialization Overhead:** The target engine reconstructs the exact file descriptor handle and maps the identical physical RAM page in microseconds.

C. LMK (Low Memory Killer) Shielding & Priority Promotion

Android's `lmkd` daemon continuously terminates background processes based on their `oom_score_adj` values. Heavy LLM runtimes are prime targets for termination:

- **Foreground Service Architecture:** Daemons invoke `startForeground(ID, notification)` bound to active notification channels.
- **Priority Clamping:** This clamps the process priority to `FOREGROUND_APP_ADJ` (OOM adjustment equivalent to an active user viewport).
- **Automated Kernel Resuscitation:** Services return `START_STICKY`, instructing the Android kernel to immediately re-instantiate the daemon if an unexpected thermal drop occurs.

D. Performance Tuning & ADB Loopback Shell Elevation

- **ADPF (Android Dynamic Performance Framework):** Background engines hook into Android's low-level Performance Hint API and Vendor Game SDKs, unlocking maximum thermal ceilings and pinning thread affinities to high-performance Kryo/Oryon CPU cores.
- **Local Wireless ADB Loopback:** Operates over `127.0.0.1:5555`. Bridges terminal commands to `adbd` running with **UID 2000 (shell) permissions**, bypassing app sandboxing to run native background binaries without root.

3. Qualcomm QAIRT, QNN & Hexagon HTP Hardware Architecture

[GGUF / Serialized Quantized Weights in Flash]



[Linux mmap() / ASharedMemory Physical DMA-BUF]



▼ FastRPC IOCTL (/dev/adsprpc-smd)

[Hexagon v79 HTP NPU Space] —► Pinned HTP0 Tensor Engine

A. Snapdragon 8 Elite Hexagon v79 Hardware Specifications

- **Throughput:** 40+ INT8 TOPs dedicated hardware acceleration.
- **FastRPC Kernel Driver:** Direct kernel interface via `/dev/adsprpc-smd`.
- **Memory Architecture:** Physical contiguous DMA-BUF / ION memory buffers registered with the DSP via `remote_register_buf()`. The NPU reads token matrices directly from physical RAM with **0% CPU copy tax**.

B. Direct On-Device Inference Without Host Compilation (GenieX)

Prior architectures required an x86 Linux computer to run Qualcomm's offline QAIRT GGUF Builder toolchain before pushing binaries via ADB.

- **The Modern GenieX Pathway:**
 - Any standard HuggingFace GGUF Q4_0 is loaded directly on-device.
 - GenieX hooks into the **Qualcomm GGML Hexagon Backend** compiled natively for Android.
 - **Zero Host Compilation:** Eliminates the x86 Linux machine requirement entirely.
- **Execution Schedulers:**
 - `--device hybrid` (Default & Fastest): Uses a per-tensor scheduler. Offloads all supported quantized operations (GEMM, activations) to the Hexagon HTP, falling back cleanly to ARM Neon CPU instructions for unsupported layers without crashing. Prefill speeds reach ~90 tok/s, decode ~27 tok/s on Qwen 3.5.
 - `--device npu`: Pins 100% of graph operations strictly to HTP0 for deterministic, jitter-free execution.
 - `--device gpu`: Routes operations to Adreno via OpenCL.
 - `--device cpu`: Pure multi-threaded ARM CPU execution.

C. Multi-Backend Quantization Formats & Context Binary Teardown

- **QNN Context Binaries (.bin):** Pre-compiled execution graphs built using `libQnnHtp.so` for Gemma 4 QAT INT4 execution.
 - **Memory Reclamation & Teardown Protocol:** When hot-swapping models, the NPU Manager executes an explicit C-level teardown:
 1. Calls `QnnContext_free()` to release DSP context handles.
 2. Invokes `munmap()` on physical DMA-BUF descriptors.
 3. Flushes HTP L2/TCM cache lines via driver `ioctl()`.
 4. Enforces a 500ms quiescence delay to ensure zero lingering locks on `/dev/adsprpc-smd` before the next model allocates memory.
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4. Repository Directory Structure

vendor-corpora/

— README.md specification)	# This document (Low-level systems architecture
— manifest.jsonl	# Master cryptographic index of all vendor specifications
— qualcomm-qairt-sdk/	# Qualcomm QAIRT, QNN & Hexagon Reference Core
— manifest.jsonl	# Local Qualcomm API index
— http-specs/	# Hexagon v79 HTP hardware limits & tensor schedules
— fastrpc-drivers/	# /dev/adsrpc-smd ioctl interfaces & buffer registries
— qnn-headers/ bindings	# libQnnHtp.so, libQnnSystem.so, and C++ header
— genieX-runtime/ flags	# GenieX GGML Hexagon backend documentation & run
— google-android-platform/	# Google Android Native Framework Core
— manifest.jsonl	# Local Android platform index
— asharedmemory-ndk/ socket code	# ASharedMemory_create, mmap, and SCM_RIGHTS
— process-isolation/ rules	# AndroidManifest.xml :process tags & isolatedProcess
— lmk-mitigation/ FOREGROUND_APP_ADJ	# Foreground Service, START_STICKY, and
— adpf-performance/ SDK hints	# Android Dynamic Performance Framework & Game

└─ wireless-adb/
protocols

adbd TCP loopback on 127.0.0.1:5555 & pairing